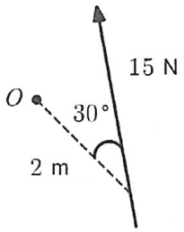
Moments: Ladders and Hinges

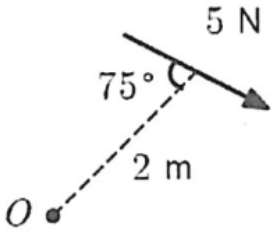
Questions 1–12

Question 1. Find the resultant moment about O in each case, giving your answer to 3 significant figures and stating whether it is anticlockwise or clockwise:

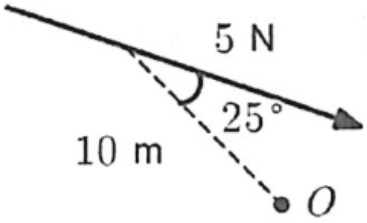
a)



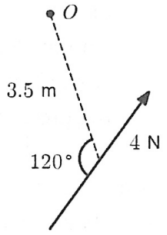
b)



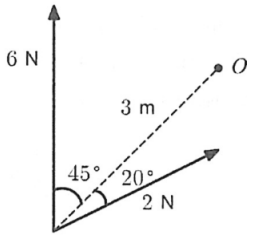
c)



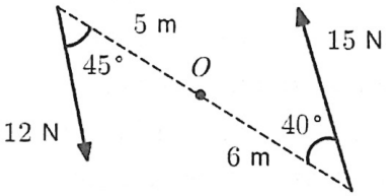
d)



e)

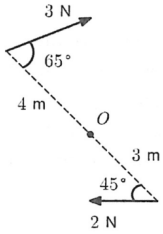


f)

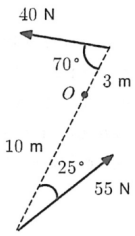


Question 1 (continued). Find the resultant moment about O in each case, giving your answer to 3 significant figures and stating whether it is anticlockwise or clockwise:

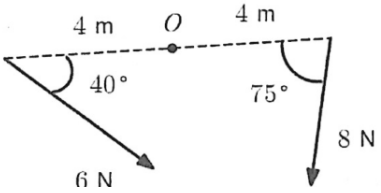
g)



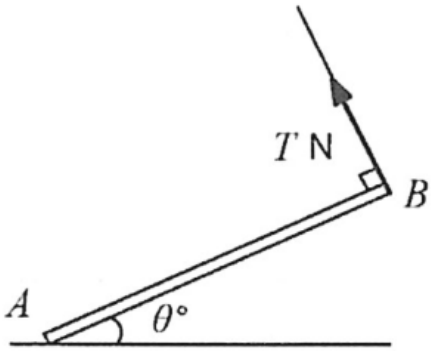
h)



i)



Question 2. The following questions use $g = 9.8$ **ms**$^{-2}$ A uniform rod AB has mass m kg and length x metres. It rests with one end A on rough horizontal ground and is held in limiting equilibrium at angle θ by a light inextensible string at B with tension T N.



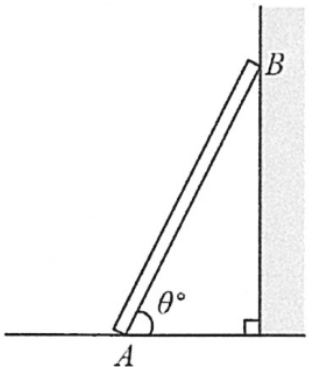
a) Draw a force diagram.

For each of the following, given

b) $m = 30, x = 10, \theta = 30$, find the value of T and the value of μ

c) $x = 8, \theta = 20, T = 180$, find the value of m and the value of μ

Question 3. A ladder modelled as a uniform rod AB with length x metres and mass m kg rests against a wall. One end, A is resting on rough horizontal ground and the other end B rests on a smooth vertical wall. The ladder is held in limiting equilibrium at an angle of $\theta°$ to the horizontal.



The coefficient of friction between the ladder and the floor is μ

a) Draw a force diagram.

For each of the following find μ, given

b) $\theta = 60, x = 5, m = 10$

c) $\theta = 45, x = 5, m = 10$

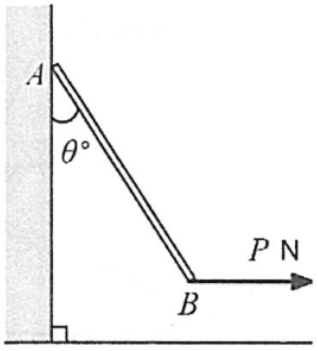
d) $\theta = 60, x = a, m = M$

A mass of $\frac{1}{2}m$ kg is loaded onto the ladder $\frac{1}{4}$ of the way up. The ladder is a metres long and has mass m kg. Find the value of μ when

e) $\theta = 60$

f) $\theta = 45$

Question 4. A uniform lever AB of mass m kg and length x metres is attached at A to a vertical wall by a smooth hinge. The acute angle between the lever and the wall is $\theta°$. The lever is held in equilibrium by a horizontal force of magnitude P N at B.



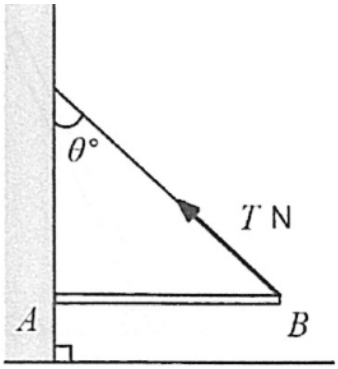
a) Draw a force diagram.

For each of the following, find P and the magnitude of the force exerted on the rod by the wall, given

b) $\theta = 30, m = 2, x = 10$

c) $\theta = 45, m = 5, x = a$

Question 5. A uniform rod AB of mass m kg and length x metres is attached to a vertical wall by a smooth hinge. It is held in place horizontally by a light inextensible string which is attached to the wall directly above A and the other end is attached to the rod at B.



The tension in the string is T N. The acute angle between the string and the wall is $\theta°$.

a) Draw a force diagram.

For each of the following, find the value of T and the magnitude of the force exerted on the rod by the wall, given

b) $\theta = 30, m = 3, x = 10$

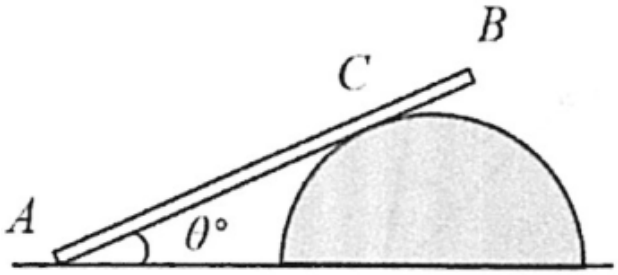
c) $\theta = 60, m = 15, x = a$

d) $\theta = 45, m = 10, x = a$

A box of mass 3 kg is added to the rod 2 metres from A. Given

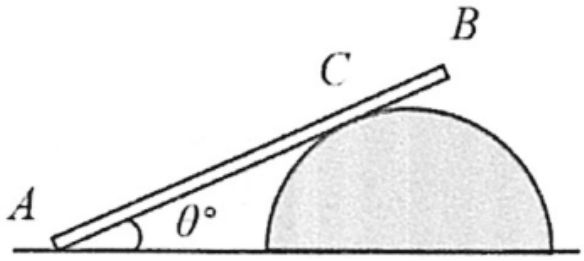
e) $\theta = 40, m = 5, x = 8$, find the value of T

Question 6. A uniform rod AB has weight 100 N and length $5a$ metres. It rests with one end A on rough horizontal ground and is resting on a smooth cylindrical drum at C where AC is $4a$ metres. The rod is angled at θ degrees to the horizontal where $\tan\theta = \frac{7}{24}$



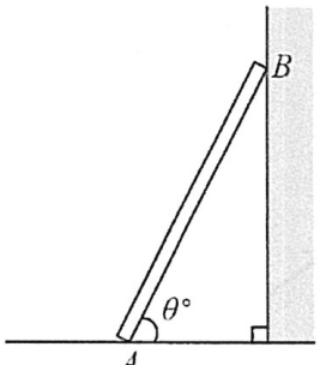
a) Draw a force diagram.
b) Find the normal and frictional components of the contact force at A.
c) Find the least possible value of the coefficient of friction between the rod and the ground.

Question 7. A non-uniform rod AB has weight 80 N which acts $2a$ metres from A. It rests with one end A on rough horizontal ground and is resting on a smooth cylindrical drum at C where AC is $4a$ metres. The rod is angled at θ degrees to the horizontal where $\tan\theta = \frac{7}{24}$



a) Draw a force diagram.
b) Find the normal and frictional components of the contact force at A.
c) Find the least possible value of the coefficient of friction between the rod and the ground.

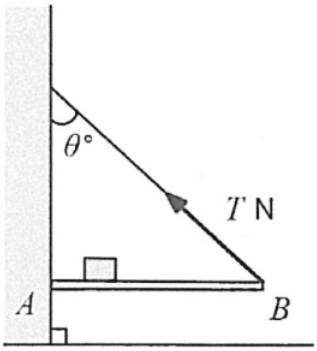
Question 8. A uniform ladder AB of length a metres rests against a wall in limiting equilibrium. One end, A rests on rough horizontal ground at an angle of θ and the other end B rests on a smooth vertical wall. The coefficient of friction is μ



Show that

$$2\mu \tan\theta = 1$$

Question 9. A uniform rod AB of mass 2 kg and length $4a$ metres with its end A against a rough vertical wall, with B being held in place horizontally by a light inextensible string which is attached to the wall directly above A

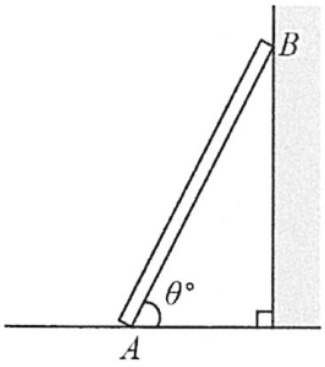


The tension in the string is T N. The acute angle between the string and the wall is $\theta°$ where $\tan\theta = \frac{3}{4}$
An object of mass 0.5 kg is placed on the rod, a metres from A

a) Find the value of T
b) Find the normal and frictional components of the contact force at A
c) Find the value of the coefficient of friction.

The following question use exact values of g

Question 10. A ladder modelled as a uniform rod AB of length $2a$ metres and mass m kg rests against a wall. One end, A is resting on rough horizontal ground and the other end B rests on a rough vertical wall. The ladder is held in limiting equilibrium at an angle of $\theta°$ to the horizontal



The coefficient of friction between the ladder and the floor is 0.4 and the coefficient of friction between the ladder and the wall is 0.1

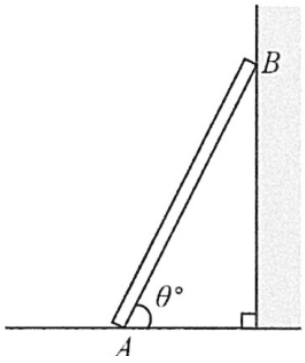
a) Show that the normal reaction at the floor is 2.5 times larger than the normal reaction at the wall.

The normal reaction at the floor is 15 N, find

b) m in terms of g

c) the value of θ to 3 significant figures.

Question 11. A uniform ladder AB of length $6a$ metres and mass m kg rests against a wall. One end, A is resting on rough horizontal ground and the other end B rests on a smooth vertical wall. The ladder is held in limiting equilibrium at an angle of $\theta°$ to the horizontal where $\tan\theta = \frac{24}{7}$



The coefficient of friction between the ladder and the floor is 0.25

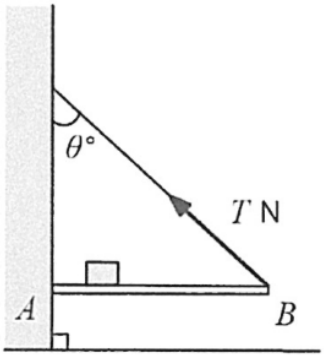
A person of mass $6m$ kg stands at the top of the ladder, whilst another person holds the ladder in place, pushing with a force of magnitude P N horizontally at A.

a) Show that the reaction of the wall on the ladder is

$$\frac{91}{48}mg$$

b) Find in terms of m and g the values of P for the ladder to remain in equilibrium.

Question 12. A uniform plank AB of mass m kg and length a metres with its end A against a rough vertical wall, with B being held in place horizontally by a light inextensible string which is attached to the wall directly above A. The acute angle between the string and the wall is $\theta°$ where $\tan\theta = \frac{4}{3}$. The tension in the string is T N.



An object of mass $5m$ kg is placed on the rod, x metres from A

a) Show that

$$T = \frac{5mg(10x + a)}{6a}$$

b) Find the % along the plank from A that the object may be placed given the maximum tension the rope can take is $8mg$ N.